Avian DNA Classification - Research Report

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Final Model Performance

model_name:	Proposed Hybrid (DNABERT+CNN+GAT+XGB)
accuracy:	0.9765
precision:	0.9909
recall:	0.9618
sensitivity:	0.9618
specificity:	0.9912
f1_score:	0.9762
roc_auc:	0.9943
pr_auc:	0.9954
mcc:	0.9534
balanced_accuracy:	0.9765
cohen_kappa:	0.953
training_time_s:	2.47
inference_time_s:	0.0

Baseline Model Comparison

Model Name	Accuracy	F1 Score	Roc Auc	Mcc	Training Time S
Logistic Regression	0.9295	0.9301	0.9699	0.8592	4.06
Random Forest	0.9758	0.9752	0.9922	0.9525	2.06
SVM	0.9743	0.9739	0.9948	0.949	19.19
KNN	0.9684	0.9675	0.9902	0.9384	0.0
Naive Bayes	0.9068	0.9001	0.9742	0.8209	0.05
XGBoost (k-mer)	0.9765	0.9762	0.9943	0.9534	4.56
Proposed Hybrid (DNA	0.9765	0.9762	0.9943	0.9534	2.47

Ablation Study

Model Name	Accuracy	F1 Score	Roc Auc
Full Model	0.9765	0.9762	0.9943
w/o DNABERT	0.9743	0.9738	0.9933
w/o GAT	0.978	0.9777	0.9929
w/o Attention	0.9765	0.9761	0.9934
w/o XGBoost (LogReg)	0.9295	0.9301	0.9699
w/o RAG (Pred-only)	0.9765	0.9762	0.9943